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CASE 13

Learning from Disappointment

Reducing the Cost of Institutional Delivery in Gujarat, India

The Case at a Glance

HEALTH GOAL: To reduce infant and maternal deaths by increasing institutional deliveries.

STRATEGY: Government payments through Chiranjeevi Yojana to private doctors to pay for normal or cesarean-section deliveries for below-poverty-level and tribal women.

HEALTH IMPACT: No evidence of a change in the probability of institutional delivery or birth-related complications in the general population between 2005 and 2010. No reduction in out-of-pocket expenditure for delivery.

WHY IT DIDN'T WORK: Poor targeting of program. Insufficient enrollment of doctors and women. Inadequate reimbursement level and persistent need for out-of-pocket expenditures. Insufficient evidence to inform better execution. Gaps in quality of care. Cultural barriers to hospital births.

FINANCING: US\$32 million from 2006 to 2012.

SCALE: More than 800 participating hospitals in Gujarat (2012).

Most of the time, babies arrive without complications—at least 85 percent of all births occur without a serious problem.¹ But when complications do arise they can be fatal, threatening both mother and infant. Most maternal death occurs during labor, during delivery, or immediately after delivery.² The World Health Organization recommends that help be close by when a woman gives birth. Ideally, a woman should give birth in an equipped facility that can provide lifesaving emergency care at the moment it is needed.³

In Gujarat, a state in northwest India, the rate of loss of mothers and babies during childbirth was unacceptably high, in part because only 55 percent of births took place in a medical facility as of 2005. This was in sharp contrast to Kerala, in southwest India, where coverage of facility births had reached 99 percent.⁴ In 2005, Gujarat's government designed and introduced a new scheme to improve birth outcomes: Chiranjeevi Yojana (CY), the "Eternal Life" scheme. It aimed to tackle the state's low rate of institutional delivery by removing one of the most important barriers for below-the-poverty-line (BPL) and tribal women: the cost of the service.

The scheme held great promise. CY was built on a strong foundation of international good practice, a vibrant private sector, and political commitment by the state's leadership. And many families benefited from CY: by 2012, the pool of roughly 800 enrolled obstetricians had delivered 800,000 babies in health facilities, including a third of all births to BPL and tribal women, the scheme's target population.⁵

Despite these impressive numbers, however, there is no evidence that CY reduced maternal or infant deaths.⁶ This is partly explained by changes in the overall health landscape: rates of institutional delivery were already rapidly increasing, and indicators of maternal and infant health were on the rise. These underlying trends reduced the relative impact of CY's contribution. In addition, some of the features of CY's design turned out to be less valuable than its planners expected.⁷ Even though improving maternal and infant health is a global health priority for which funding is at an all-time high, the CY experience serves as a cautionary tale for programs that seek to achieve this goal by increasing institutional delivery.⁸

This case was originally authored by Miriam Temin.

The Toll of Maternal and Infant Mortality

Worldwide, mothers and babies today are more likely to survive delivery than their forebears in decades past. In India, a lower-middle-income country, maternal and child health has improved significantly in recent years. Maternal mortality dropped from 212 deaths per 100,000 live births in 2007 to 178 in 2012. In the same period, annual infant mortality declined from 55 to 42 deaths per 1,000 live births.⁹ But not surprisingly, these health gains have not been evenly distributed, and mortality rates vary significantly by socioeconomic status and by state.

Gujarat State in northwestern India is one of the five wealthiest states in the country.¹⁰ Its 60 million residents benefit from an economy that is growing by nearly 10 percent per year.¹¹ Indians view the government of Gujarat as progressive when it comes to health, and its representatives have become national advocates for insurance coverage for the poor.¹²

This leadership paid off in statewide maternal and infant mortality statistics that looked better than the national averages. For example, from 2010 to 2012, 122 women died annually in childbirth per 100,000 live births in Gujarat, compared with 178 nationwide.¹³ One reason for better maternal health was the rising number of institutional deliveries: the average proportion of women delivering in institutions more than doubled from 2001 to 2010, from 40.7 percent to 89.3 percent, a significant and durable shift away from home births.¹⁴ CY contributed to these improving statistics. Yet despite the program, maternal health has improved at a slower pace in Gujarat than in India as a whole.¹⁵

Putting an Idea in Motion: Gujarat Smooths the Path to Institutional Delivery

Facility-based care during labor and delivery is a global recommendation. While there is no guarantee that all possible problems will be successfully dealt with in a hospital or clinic, facilities that provide maternity care are more likely than other places to have the trained staff, infrastructure, and protocols necessary to manage most complications.¹⁶ Children delivered in a facility also benefit in secondary ways; for example, they are more likely than those born at home to receive their vaccinations and be breast-fed.¹⁷

In India, the National Rural Health Mission was the government's flagship program to widely increase access to healthcare, put in place in 2005.¹⁸ Shifting all deliveries into institutions was a national objective under this program. India is one of two countries (Nigeria is the other) that account for a third of all maternal deaths in the world, making this an urgent priority.¹⁹

Despite the national objective, India's public health-care system lacked the capacity to cope with all pregnant women.²⁰ Human resource distribution problems were common, and emergency obstetric care was sparse. For example, in Gujarat's rural areas, a mere seven or eight government obstetricians served a population of 32 million.²¹ Women in Gujarat had a difficult time reaching government facilities, and if and when they arrived the quality of care was low. Private doctors, in contrast, were more readily available. Three-quarters of all the doctors in Gujarat worked in the private sector,²² and 60 percent of all deliveries in Gujarat took place in private facilities by 2010.²³ The shift to private care also signaled women's widespread lack of trust in public facilities and their perception that private care was better.²⁴

In India, healthcare is provided and funded at the state level. It was up to the most powerful politician in Gujarat, Chief Minister Narendra Modi, to fix problems with the health sector in his state. Modi made clear that his government was open to working with either public or private providers to do so—whichever could do a better job.

Modi wanted to make specific changes that would ensure the health of mothers and babies.²⁵ His government recognized that private providers were available to pregnant women where public providers were not. The state government designed a scheme that leveraged the vibrant private sector by incentivizing it to serve the subpopulations of women that had the highest risk of maternal mortality: poor and tribal women. The government aimed to do this by paying private providers to deliver babies for women who possessed a BPL or tribal designation. By 2005, the state government was ready to launch CY in five disadvantaged pilot districts with the support of civil society organizations and the German development organization Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ).²⁶ Following the reported success of the pilot, CY expanded statewide in 2007.

Chiranjeevi Yojana in Action

The design of the scheme was relatively straightforward: the government recruited private obstetricians, contracted them to deliver babies for tribal and BPL women, and paid them for the deliveries and associated costs. Community leaders, local obstetric and gynecological societies, and district health teams helped attract eligible doctors.²⁷ Doctors' enrollment criteria were clear: ownership of basic facilities; at least 15 beds; and access to blood, anesthesiologists, and emergency surgery.²⁸

Obstetricians could claim reimbursement for eligible deliveries on a monthly basis. For every 100 deliveries, the District Programme Management Unit paid them a fixed amount. A leading nongovernmental organization, Society for Education Welfare and Action–Rural (SEWA Rural), calculated the payment based on average expenses for 100 routine and complicated deliveries;²⁹ the reimbursement rate was finalized following consultations with important stakeholders, for example obstetric and gynecological societies.³⁰ The reimbursement rate increased over the years, reaching INR380,000 for every 100 deliveries by 2013 (roughly US\$69 per delivery in 2013).³¹ The scheme covered only the costs of delivery, not prenatal or postnatal care. Part of the reimbursement amount was intended to go to the women for transport and food, plus a small incentive payment for their escorts.³²

The government hoped that the program would attract BPL and tribal women—groups with higher rates of home deliveries and maternal complications than other women. The potential beneficiaries had to show proof of eligibility—a BPL or tribal card—to be treated by a CY doctor under the scheme.³³ Community health workers were charged with informing women of their eligibility.

The Payoff: No Difference for Maternal Health

After observing low uptake of CY, in 2009 the government commissioned an independent evaluation from the Collaboration for Health System Improvement and Impact Evaluation in India (COHESIVE-India). The evaluation's findings, since corroborated by other studies, came as a surprise: many women did shift from home births to institutional delivery over this period, but CY did not appear to accelerate the shift.³⁴ In other words, there was no evidence that the scheme significantly

increased the likelihood of institutional delivery for women in the target group,³⁵ nor for women in the broader population.³⁶

The increase in facility deliveries and the improving health of Gujarat's women and babies was great news; however, the evaluation indicated that CY had little to do with it. Consistent with the finding that CY did not significantly affect access to institutional delivery, the evaluation also failed to find a significant effect of the scheme on indicators of maternal and infant health in the general population (see Box 1). The researchers' analysis suggested that 54 percent of all women statewide had delivery complications after CY's launch—and the scheme did not reduce this probability.³⁷ It is worth noting that the analysis assessed outcomes for all pregnant women statewide, not the poorer women in the scheme's target population; with the evidence available, it is not possible to determine if this influenced the study's conclusions.

Notably, CY coincided with an increased rate of cesarean sections among BPL women, according to another study, although researchers could not conclusively confirm whether these were needed procedures.³⁸ CY was actually designed to disincentivize unnecessary cesarean sections through a standard per-delivery reimbursement rate—that is, doctors were paid the same amount per 100 deliveries whether or not they performed C-sections. This may have actually reduced the likelihood that doctors would perform a C-section on a CY woman even when medically necessary, and would instead refer the complicated cases elsewhere to avoid the medical risks and costs.³⁹

CY's founders were convinced that lower financial barriers to facility-based delivery would improve outcomes for new mothers and their babies. The cost barrier was real: Indians' out-of-pocket payments for healthcare are among the highest in the world.⁴⁰ The average household expenditure for a normal private delivery in India is US\$84, an impossible sum for many in CY's target population.⁴¹ Unfortunately, out-of-pocket expenditures for delivery held steady over five years of CY implementation.⁴² News articles shed light on one explanation: some participating doctors offered eligible women additional services beyond those covered by CY with no discernible benefits for health.⁴³ Studies have confirmed these reports, also finding that some doctors pocketed the portion of the CY payment intended for transport and food reimbursement.⁴⁴

Box 1. Strength of the Evidence⁴⁵

The government of Gujarat invited the Collaboration for Health System Improvement and Impact Evaluation in India (COHESIVE-India) to evaluate Chiranjeevi Yojana in 2009. Using the phased introduction—first in five northern districts and then in the remainder of the state—researchers conducted a retrospective study with the later districts serving as controls for the first wave.

Researchers used population-level data to account for bias in the self-selection of pregnant women into facility births. They also triangulated their results with analysis of district-level household and facility survey data. Their findings were consistent across the two different data sources: the program showed no significant impact on the probability of institutional delivery or on birth-related complications.

Their conclusions must be interpreted with a number of limitations. First, the initial five districts differed from the rest of the state in important ways—residents were more rural, poorer, and less literate—which may have influenced the results. Second, changes in maternal mortality impact are notoriously difficult to measure because maternal deaths are relatively infrequent, leading to statistical uncertainty around

the evaluation's study results. Third, the study relied on mothers' self-reported recollections about whether they experienced complications, which are often biased.⁴⁶ Finally, the evaluators compared their sample not only with the poorer women in the target group but also with the entire population of women in Gujarat. It is possible that this wide-lens view may have obscured an effect for the smaller subsample.

The COHESIVE-India evaluation is not the only word on this program; indeed, its findings contradict several previous studies of the same program. The different study designs explain at least some of these discrepancies and suggest that the COHESIVE-India results are likely to be more accurate than those found in earlier studies. Of particular note in previous studies is the use of before-and-after comparisons, which made it hard to distinguish CY's impact from other ongoing trends—most importantly, the already increasing rate of institutional delivery before program implementation. Yet another large evaluation, funded by the European Union, is ongoing, poised to yield further insights into CY's performance and impact.

At What Price?

Funding for CY came from central government coffers through the National Rural Health Mission.⁴⁷ At CY's inception in 2006, the government of Gujarat paid providers INR1,850 (US\$40) for every delivery to beneficiaries, gradually increasing the reimbursement to INR3,800 (US\$69) by mid-2013. In total, the state government paid out US\$32 million to participating hospitals between 2006 and 2012.

CY's funding came via three different sources, which made it difficult to track. This may have accounted for underspending: as of early 2010, news reports alleged that 36 percent of the total project budget remained undisbursed.⁴⁸

At the same time that the CY allocation grew, state funding for health also expanded. According to state government sources, the state budget allocated for health increased in both absolute and proportionate terms, from 3.1 percent to 4.2 percent of the total state budget between 2005 and 2010. This reflected a total state health budget of INR11.55 billion (US\$266 million) in 2005 and INR29.88 billion (US\$434 million) in 2010.⁴⁹

Why Chiranjeevi Yojana Fell Short in Gujarat

CY was well conceived: it had a credible theory of change, leveraged a vibrant private sector, and benefited from high-level political commitment.⁵⁰ So what explains the limited impact? A number of issues appear to have constrained CY, including targeting, financing, coverage, and management. The quality of private maternity care and social determinants that make home births attractive also limited the program's impact. However, it is impossible to fully appreciate the role of these factors without more qualitative data, underscoring the need for more studies that use a range of methods.

The scheme financed delivery for an enormous number of women, but were they the right women? Evidence suggests no—that CY struggled to reach its target population.⁵¹ Some of the women who shifted into CY were already using private healthcare, while other eligible women continued delivering at home. Part of the problem was identifying the right women. As is common in India, CY relied heavily on BPL cards to determine eligibility. In theory, households receive the cards when they score 20 or below on an index of multidimensional poverty.⁵²

In practice, around 40 percent of eligible households in Gujarat do not have a BPL card, and around 40 percent of BPL card holders do not meet the official eligibility criteria.⁵³

Another targeting challenge related to geographic coverage. The distribution of CY-enrolled doctors had a distinctly urban bias, and women in some rural areas were less aware of the scheme than women elsewhere.⁵⁴ This is disappointing because the scheme emphasized rural penetration—the health commissioner and director even conducted personal visits to convince rural doctors to join.⁵⁵ Yet CY uptake was higher in the more rural districts, suggesting some rural women found their way to participating urban doctors.⁵⁶

The government planned that CY reimbursements would be sufficient to cover all delivery costs, from medical care to food to transportation. Yet despite several increases over the scheme's lifespan, doctors complained that the reimbursement level was too low to cover these expenses. This affected CY in several ways.

First, the reimbursement level influenced who joined: only 740 of the 2,000 private obstetricians in the state had signed up by 2006.⁵⁷ The scheme initially attracted younger doctors with new medical practices; these providers seized on CY as a way to build their client base.⁵⁸ And some doctors saw CY as a charitable undertaking rather than a true public-private partnership. Altruistic early adopters signed up,⁵⁹ but the lack of financial incentive likely deterred many doctors.

Second, the reimbursement level likely played a role in provider attrition. The number of participating obstetricians grew to 870 in 2008, but then fell to 660 by 2010, exacerbating CY's inadequate coverage.⁶⁰ As of 2011, 40 percent of subdistricts had no participating providers.⁶¹ Some early adopters may have left as the demand for CY deliveries increased and their nonreimbursed expenses grew, eroding their bottom line.⁶² Notably, most attrition came from large cities, where some doctors feared that a poorer client base would damage their reputations.⁶³ Rural doctors, who faced less competition and were less concerned about stigma associated with lower-status clients, may have been a better fit for the scheme. Some doctors who stuck with CY were those who would otherwise have had trouble attracting clients, perhaps because they provided low-quality care.

Third, the low reimbursement level made the scheme more of a subsidy for women than a ticket to free care. Women continued to face transportation costs, informal payments, and other expenses.⁶⁴ Some doctors—respond-

ing to insufficient reimbursement—passed on their costs to women, either by charging them for other services or by keeping the amount earmarked for them. A combination of these factors kept demand for CY low. Because families still had to pay for delivery, they may have decided that the benefit of receiving private maternity care from a participating doctor—potentially an unpopular one—failed to outweigh the cost.

Another challenge was that CY managers lacked solid evidence to inform their planning. While consultations with professional organizations, providers, and other stakeholders had preceded the launch, there was no comprehensive formative research on the barriers to institutional delivery or the quality of private maternity care to inform the design.⁶⁵ Insufficient monitoring remained a problem throughout, leaving unanswered questions about quality, referrals, and details of the deliveries.⁶⁶ Furthermore, the payments to doctors were based on self-reports, which were not checked or followed up.⁶⁷ Participating doctors had no incentive to report accurately and may have falsified records to claim a higher reimbursement.⁶⁸

Implications for Global Health

CY was the first large-scale public-private partnership to increase institutional delivery in a low- or middle-income country, and it was widely viewed as a success, even receiving the *Wall Street Journal* Asian Innovations Award in 2006.⁶⁹ Based on the CY experience, the government of Gujarat launched other insurance schemes for the poor. Mukhyamantri Amrutum Yojana (MAY), for example, provides up to US\$3,400 per family to treat critical illness in a government facility.⁷⁰ Yet research on CY has yielded contradictory results, highlighting the implications of study design and the paramount importance of experimental or quasi-experimental research, supplemented by qualitative inquiry.

CY shares features with other programs that provide vouchers and eliminate user fees, either for specific subgroups or entire populations (described elsewhere in *Millions Saved*). The removal of health user fees is an increasingly popular policy choice and an important step toward universal health coverage. CY's experience resonates elsewhere, for example, with India's Janani Suraksha Yojana (JSY) to promote institutional births—the largest conditional cash transfer program in the world and a central spoke in the National Rural Health Mission.

While JSY has increased uptake of public-sector deliveries, it too faces controversy about its health impact.⁷¹

Eliminating user fees, however, is not sufficient on its own to improve health outcomes; other barriers to accessing healthcare persist in health systems, communities, and households. Social determinants, for example, profoundly influence access to health services, warranting serious consideration in programs to equitably improve maternal health.⁷² Gender inequality also shapes women's access to healthcare, and their low social position limits their control over their own health during pregnancy. Among one sample of CY beneficiaries, none of the women chose where they would give birth; rather, auxiliary nurses, midwives, husbands, and mothers-in-law made the decision on their behalf.⁷³ Empowering women is a clear priority, but the route to institutional delivery also must run past their families and trusted community health workers.

Other noncost barriers must be addressed as well, including the reasons that women in Gujarat continue to deliver at home: the convenience of home delivery, fear of health facilities, and the difficult trip to the hospital.⁷⁴ The powerful norms that influence maternity care decisions fell far outside of the scope of CY.

With the ascendance of former state secretary Modi to prime minister in 2013, elements of the CY design may show up elsewhere in India. Gujarat's experience reveals the importance of carefully considering many different components of program design, including financing, monitoring, entry criteria, and quality assurance, alongside the social determinants of health. And as India works to achieve health equity, lessons from CY, JSY, and other demand-side health financing approaches must inflect the next generation of financing programs to increase health equity.

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